REGRESSION AUTOPSY

Eight Ways Your Model Is Lying to You

Complete Build Specification

For Implementation by Claude Code

CONFIDENTIAL BUILD DOCUMENT

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1. Course Overview & Core Promise

> *Regression is the most dangerous tool in data science. Not because it doesn't work — because it almost works. It gives you a number, a confidence interval, a p-value, and a clean summary table. Everything looks authoritative. And in most observational analyses, the number is wrong. This series of notebooks will show you exactly how, exactly why, and exactly what to do about it.*

Course Title: Regression Autopsy: Eight Ways Your Model Is Lying to You

Format: Eight core interactive Colab notebooks + one optional deep dive

Core Promise: By the end of this course, you will never blindly trust a regression again — and you'll know exactly when you should trust one.

Emotional Arc: Trust → Betrayal → Understanding → Mastery → Redemption

The student begins by trusting regression blindly (Notebook 1), gets burned six different ways (Notebooks 1–6), learns to diagnose and mitigate each problem, confronts the limits of mitigation with real data (Notebook 7), and discovers that regression can be trustworthy when paired with a research design that earns that trust (Notebook 8).

Opening Framing (displayed in Notebook 1, Cell 0): "You'll work with one company's data across eight notebooks. Each notebook will ask you to make a judgment. Each time, you'll be wrong — but in a precisely predictable way. By the end, you'll understand not just how to run a regression, but when to trust one."

2. Target Audience

Primary: Anyone who runs regressions without formal econometrics training — data analysts at startups, product managers interpreting A/B tests, policy researchers, journalists, engineers.

Secondary: Instructors who want a ready-made module for statistics, econometrics, or data science courses.

Language Principles

- Lead with the consequence, not the mechanism. "Your number is too big, here’s why" before the formula.
- Theorem names appear in footnotes or collapsible cells, never in the main narrative flow.
- Every concept has a one-sentence version before the formal version.
- The insight is complete without the math. The math is there for those who want it.

3. Pedagogical Framework

3.1 The Six-Beat Structure

Every notebook follows the same six-beat arc. This consistency builds familiarity — by Notebook 3, the student knows the rhythm and can focus on the content rather than the structure.

Beat	Name	Purpose	Student Experience
1	Trap	Student commits to a wrong answer via interactive widget	Confidence: "I know this"
2	Reveal	One visualization exposes the problem	Shock: "Wait, what?"
3	Formalize	Theorem stated in LaTeX, confirmed by Monte Carlo simulation	Understanding: "Oh, that's why"
4	Destroy	Interactive sliders let the student control violation severity	Exploration: "What happens if I…"
5	Repair	Toggle activates the fix, side by side with the broken version	Relief: "There's a solution"
6	Limit	Second slider finds the boundary where the repair fails	Wisdom: "But it's not a silver bullet"

3.2 The Trap Widget System

The trap is the most critical element. Its purpose is to make the student commit to an answer before being corrected. This commitment is what makes the correction memorable.

Design Requirements for Traps

- Use ipywidgets radio buttons or checkboxes, not free text, so responses are machine-readable.
- Include a Submit button. The next cell's output is gated until submission.
- Traps must be hard enough to catch experienced practitioners, not just beginners. For example: ask the direction of bias, not just whether bias exists.
- Record the student's response in a JSON file for the final dashboard in Notebook 8.
- Gate system is pure Python (ipywidgets), no JavaScript injection. Must work in Colab, Jupyter, JupyterLab, and VS Code.

3.3 Recurring Dataset

All eight notebooks use the same company (NovaMart) and the same variables. The student builds a relationship with this data across notebooks. When a coefficient changes from one notebook to the next, they remember the old value. The recurring dataset creates emotional continuity that isolated examples cannot.

3.4 Discussion Prompts

After each Destroy beat, a markdown cell poses a discussion question designed to provoke argument between two students or serve as a reflective prompt for solo learners. These have no single correct answer — they surface the judgment that statistics requires beyond mechanics.

3.5 Lingering Questions & Transition Cards

Each notebook ends with a lingering question that the next notebook answers, plus a styled HTML card with a one-click Colab link to the next notebook. This is the Netflix-style "next episode" prompt delivered at the moment of maximum engagement.

4. Visual Design System

4.1 Color Palette

Consistent across all eight notebooks. No exceptions. No notebook-specific color schemes.

Color	Hex	Meaning	Usage
Blue	#2E5EA8	OLS estimator	OLS distribution, regression line, OLS confidence intervals
Gold	#D4A017	Truth	True parameter value (vertical line), true DGP
Red	#C0392B	Bias / Violation / Error	Biased distributions, violated assumptions, missed CIs
Green	#27AE60	Repair working	Corrected estimator, restored coverage
Gray	#7F8C8D	Alternative estimators	Non-OLS estimators in Gauss-Markov comparisons

4.2 Accessibility: Secondary Visual Channels

Every color choice must be paired with a secondary channel so the visualization works in grayscale and for colorblind users.

Element	Color	Line Style	Marker
OLS	Blue	Solid	Circle
Truth	Gold	Thick solid	Diamond
Violation / Bias	Red	Dotted	Triangle
Repair	Green	Dash-dot	Square
Alternatives	Gray	Dashed	None

4.3 Interactive Elements

- Sliders: horizontal, at the top of the cell. Left end = assumption holds (green label). Right end = assumption maximally violated (red label). Default position always left. A thin colored bar fading green-to-red beneath the slider.
- Dynamic titles: a sentence above each plot that updates with slider position. Format: "OLS is unbiased ✓ (bias = 0.00)" or "OLS is biased ✗ (bias = 2.47, predicted: 2.51)".
- Two-column layout: visualization on the left, live-updating summary panel on the right showing current parameter values, theoretical prediction, simulated result, and discrepancy.
- Slider positions: discrete, snapped values. No continuous sliding. Prevents interpolation artifacts. Slider for ρ has 21 positions: -0.95, -0.85, …, 0.85, 0.95.

4.4 Sidebar Visual Treatment

Sidebar plots use the same hues as the main flow but at 60% saturation. This visually signals supplementary status. Sidebars are collapsed by default using Colab's #@title feature with a provocative label.

5. Technical Infrastructure

5.1 Shared Module: company_sim.py

Embedded in each notebook's collapsed setup cell. No cross-notebook imports. No network dependencies. The notebook must work offline.

CompanySimulator Class (~80 lines)

- Constructor takes pathology strength parameters with realistic defaults.
- generate(n=500, seed=42) → pandas DataFrame with columns: revenue, ad_spend, staff_count, satisfaction.
- truth() → DataFrame adding demand_U column + dictionary of true parameters.
- dgp_summary() → LaTeX string of the full specification.
- Pathology methods: set_heteroscedasticity(strength), set_endogeneity(strength), set_nonlinearity(curvature).
- Data generation order: U first, then X_1 and X_2 from U, then Y from X_1, X_2, U, ε, then X_3 from Y. X_3 is post-treatment — never in the true DGP for Y.

MonteCarloEngine Class (~40 lines)

- run(estimator_fn, param_grid, n_reps=5000) → precomputes draws across parameter grid, returns numpy array.
- Stores results in dictionary keyed by parameter grid values.
- Slider callbacks index into the array. Response time < 50ms.
- quick_run(estimator_fn, dgp_fn, n_reps=1000) → single-configuration simulation for sidebar mini-sims. No caching.
- Precomputation runs during setup with progress bar. Estimated time: 20–30 seconds on free Colab.

DiagnosticSuite Class (~30 lines)

- Single function call producing four-panel residual diagnostic: residuals vs. fitted, QQ plot, scale-location, leverage.
- Plus VIF table, Breusch-Pagan test, Durbin-Watson statistic, Jarque-Bera test.
- Consistent visual format across all notebooks. Builds familiarity by Notebook 7.

AutopsyReport Widget Factory (~30 lines)

- Notebooks 3–6: lightweight two-question version ("What is the biggest threat?" and "How would you check?").
- Notebook 7–8: full sensitivity analysis version with four fields + contour plot.

- Third field added from Notebook 3 onward: "Which real-world disaster is most analogous?" with dropdown of previously encountered sidebars.
- Responses stored to JSON for the final dashboard in Notebook 8.

5.2 Gate System (~20 lines)

Module-level dictionary _trap_responses = {}. Trap widget's submit callback writes to this dictionary and saves as JSON to /content/trap_log.json. Reveal cells check for the key. If missing, display warning. Pure Python. No JavaScript. Platform-independent.

5.3 Dependencies

All pre-installed on Colab. No pip installs required: numpy, pandas, matplotlib, statsmodels, ipywidgets, IPython.display. For Notebook 7: the Lalonde dataset is base64-encoded in the setup cell. Zero external data dependencies.

5.4 Precomputation Strategy

Every slider snaps to discrete values. Each position has exactly 5,000 precomputed Monte Carlo draws stored in a dictionary. Slider movement triggers a dictionary lookup, not a computation. Response time under 50 milliseconds. Precomputation runs in setup cell with progress bar. Front-load engagement: the scenario and trap question appear before precomputation, so the student reads and answers while the engine runs in a subsequent cell.

6. The Data Generating Process (DGP)

A single coherent DGP underlies all eight notebooks. Every pathology is present simultaneously in one business story. Each notebook isolates and examines one pathology.

6.1 The Business Story

NovaMart is an online retailer. The outcome Y is monthly revenue (thousands). Predictors: ad spend X_1 (thousands), number of sales staff X_2, and customer satisfaction score X_3. A hidden variable U represents market demand, which the student never observes directly.

6.2 Formal Specification

$$Y = 50 + 8 \cdot \log(X_1) + 3 \cdot X_2 + 2 \cdot U + \varepsilon$$

Where:

- $U \sim N(0, 1)$ — unobserved market demand
- $X_1 = 20 \cdot U + \eta_1$ — ad spend driven by demand (creates endogeneity)
- $X_2 = 5 \cdot U + \eta_2$ — staff count driven by demand
- $X_3 = 0.1 \cdot Y + \eta_3$ — satisfaction caused by revenue (post-treatment / bad control)
- $\varepsilon \sim N(0, 0.5 \cdot X_1)$ — heteroscedastic errors (bigger campaigns = more variance)
- $\eta_1, \eta_2, \eta_3 \sim N(0, \sigma^2)$ — independent noise terms

6.3 Pathologies Present

Pathology	Source in DGP	Notebook
Omitted variable bias	U drives both X_1 and Y; omitting U biases β_1 upward	1
Heteroscedasticity	ε variance scales with X_1; big campaigns have unpredictable returns	2
Inflated significance	Large n (50,000 store-months) makes trivial effects significant	3
Misspecification	True relationship is $\log(X_1)$, not linear X_1	4
Overfitting	Adding polynomial terms improves training fit, destroys test fit	5
Endogeneity / Reverse causality	X_3 is caused by Y; including it as control creates collider bias	6

All simultaneously	Lalonde real data replaces simulated data	7
None (by design)	RDD with valid design eliminates confounding	8

6.4 Data Generation Order

Critical: generate in causal order to avoid simultaneity. Step 1: Generate U, η_1, η_2, ε. Step 2: Compute $X_1 = 20U + \eta_1$ and $X_2 = 5U + \eta_2$. Step 3: Compute $Y = 50 + 8 \cdot \log(X_1) + 3 \cdot X_2 + 2 \cdot U + \varepsilon$. Step 4: Compute $X_3 = 0.1 \cdot Y + \eta_3$. Step 5: Drop U from the observed dataset. The student sees only revenue, ad_spend, staff_count, satisfaction.

6.5 Collider Bias Note

X_3 (satisfaction) is NOT in the true DGP for Y. It is caused by Y. Including X_3 as a control is a collider conditioning error. In Notebook 1's repair cell, toggling X_3 into the model should make the coefficient worse, not better. This is the only place in the course where a DAG is shown (three nodes: Ad Spend → Revenue → Satisfaction, with Demand → Revenue).

7. Theorem Map

Each notebook is anchored by one core theorem. The theorems form a logical chain where each generalizes the previous one. The student who reads the formal statements in sequence sees a connected structure, not isolated results.

NB	Theorem	Formal Statement	Dependencies
1	Omitted Variable Bias	$E[\hat{\beta}_1] = \beta_1 + \beta_2\delta_1$ where $\delta_1 = Cov(X_1, X_2)/Var(X_1)$	None
2	Variance Decomposition	Homoscedastic: $Var(\hat{\beta}) = \sigma^2(X'X)^{-1}$. Heteroscedastic: $(X'X)^{-1}X'\Omega X(X'X)^{-1}$	NB 1
3	t-Statistic Anatomy	$t = \hat{\beta}/SE(\hat{\beta})$. Power = $P(\|t\|>c \mid \beta\neq 0)$. Increases with n and effect size.	NB 2
4	Specification Error	If $Y=f(X)+\varepsilon$ and we fit $Y=X\beta+\varepsilon$, OLS converges to linear projection, not $f'(x)$.	NB 1–3
5	Bias-Variance Tradeoff	$E[(\hat{y}-y)^2] = Bias^2 + Variance + \sigma^2$. Adding parameters reduces bias, increases variance.	NB 4
6	IV Consistency	If $E[\varepsilon X]\neq 0$ but $E[\varepsilon Z]=0$, $E[ZX]\neq 0$: $\hat{\beta}_IV = (Z'X)^{-1}Z'Y$ is consistent.	NB 1
7	Sensitivity Analysis	Robustness value RV = minimum partial R^2 to nullify conclusion. Cinelli-Hazlett.	All
8	RDD Identification	At cutoff c, if $E[Y_0\|X=c]$ continuous, gap = $E[Y_1-Y_0\|X=c]$.	NB 4, 5
1.5	Frisch-Waugh-Lovell	$\hat{\beta}_1$ from $Y\sim X_1+X_2$ = $\hat{\beta}$ from $M_2Y \sim M_2X_1$ where $M_2 = I-X_2(X_2'X_2)^{-1}X_2'$	NB 1

Meta-lesson: each theorem generalizes the previous. OVB is a special case of endogeneity. Classical variance is a special case of the sandwich estimator. The linear model is a special case of all models. Relaxing the assumptions that justified the previous notebook's conclusion generates the next notebook's theorem.

8. Notebook Specifications

Each notebook is specified with its full cell structure, trap design, interactive elements, and connections to other notebooks.

8.1 Notebook 1: Why Your Coefficient Is Wrong

Core concept: Omitted variable bias

Core message (plain language): "If something you didn't measure is correlated with both your input and your output, your coefficient is wrong. Not noisy. Not uncertain. Wrong. Systematically shifted in one direction."

Estimated time: 45 minutes

File: 01_coefficient.ipynb

Cell-by-Cell Specification

Cell 0 — Setup (collapsed, #@title): Contains CompanySimulator, MonteCarloEngine, AutopsyReport, gate system, all imports. ~200 lines. Runs precomputation: 5,000 Monte Carlo draws at each of 21 ρ values (-0.95 to 0.95). Progress bar. ~20 seconds on free Colab.

Cell 1 — The Hook (markdown): Provocative opening paragraph ("Regression is the most dangerous tool..."). Course framing. Then scenario: "You're a data analyst at NovaMart. VP of Marketing wants to know: does ad spend drive revenue? She's deciding whether to double the budget. You have 500 months of data across regional offices."

Cell 2 — The Trap: Runs simple regression: revenue on ad_spend only. Displays statsmodels summary. Widget asks: "Based on this regression, what do you conclude?"

Options:

- A: "Ad spend strongly increases revenue (coefficient ≈ 4.7, $p < 0.001$). Recommend doubling the budget."
- B: "There's a positive relationship but I'm not confident it's causal."
- C: "The coefficient is biased upward — the true effect is smaller."
- D: "The coefficient is biased but I can't determine the direction."

Correct answer: C. Option A is the naive answer. Option D is the sophisticated-but-wrong answer (the direction IS determinable). Submit button gates Cell 3.

Cell 3 — The Reveal: Displays recorded answer. Scatterplot of revenue vs. ad_spend, points colored by hidden U (demand). Colorbar: blue = low demand, gold = high demand. Pattern visible:

high-demand periods cluster upper-right. Formal OVB formula in LaTeX: $E[\hat{\beta}_1] = \beta_1 + \beta_2\delta_1 = 2.3 + 2.0 \times 1.2 = 4.7$. Tailored response based on which option the student chose.

Cell 4 — Monte Carlo Confirmation: 5,000 replications with confound present. Animated histogram building particle by particle (200 frames, 25 particles/frame, 8 seconds). Distribution centers on 4.7. Gold vertical line at true value 2.3. Red annotation: "Bias = 2.4". Text confirms: "OVB formula predicts 4.7. Simulation mean is 4.698. Theory confirmed." Skip Animation button available.

Cell 5 — The Destruction Playground: Slider: "Correlation between demand and ad spend (ρ)" from -0.95 to 0.95 in steps of 0.1. Default: 0.0 (no confounding). Left panel: histogram of 5,000 $\hat{\beta}_1$ estimates from precomputed cache. At ρ=0 centered on truth (2.3). Slides as ρ changes. Right panel: live summary with ρ, theoretical bias, simulated mean, true β_1. Dynamic title updates. Near-perfect match between predicted and simulated bias at every position.

Cell 6 — Discussion Prompt (markdown): "Your colleague says: ‘I controlled for every variable in the dataset so there can’t be omitted variable bias.’ What’s wrong with this reasoning?"

Cell 7 — The Repair: Same layout as Cell 5. Toggle button: "Include demand (U) as control: ON/OFF". When ON, histogram snaps to center on 2.3 regardless of ρ. Second toggle: "Include customer satisfaction (X_3) as control". When ON instead of U, coefficient gets worse. Dynamic title turns red with warning. DAG rendered with matplotlib: three nodes (Ad Spend → Revenue → Satisfaction) + (Demand → Revenue). Red X when bad control active. One-sentence disclaimer: "This is a causal diagram. We won’t use these formally, but this picture explains why controlling for satisfaction backfires."

Cell 8 — The Limit of Repair: Slider: "Proxy quality: 0% to 100%". At 100% the proxy perfectly measures U; bias eliminated. At 50%, noisy proxy gives partial debiasing. At 0%, full bias returns. Smooth curve showing residual bias as function of proxy quality. Live readout: "With X% proxy quality, residual bias = [value]. You’ve eliminated Y% of the confound."

Cell 9 — Disaster Sidebar (collapsed): Title: "📎 Real-World Disaster: The hormone therapy that killed". Tabbed widget with three tabs: The Story (HRT narrative, 3 paragraphs), The Math (OVB formula: 0.5 + (-3.0)(0.6) = -1.3; sign is wrong), The Simulation (1,000 datasets, distribution centered on -1.3 with gold line at true +0.5).

Cell 10 — Mini Autopsy: Two text input fields: "What is the biggest threat to this estimate?" and "How would you check if that threat is real?" Responses stored to JSON.

Cell 11 — The Bridge (markdown + HTML): Lingering question: "Your coefficient is now correct. But can you trust the standard errors?" Netflix-style card linking to Notebook 2.

8.2 Notebook 2: Why Your Uncertainty Is Wrong

Core concept: Heteroscedasticity and standard errors

Core message: "Your coefficient might be right but your margin of error is wrong. You think you're 95% sure. You're actually 82% sure."

Estimated time: 30 minutes

File: 02_uncertainty.ipynb

Trap: Student includes correct controls from NB1. Coefficient is correct. CI shown as [6.2, 9.8]. Trap asks: "Are you confident the true effect is in this range?"

Reveal: 1,000 Monte Carlo confidence intervals stacked vertically. True value as vertical line. ~18% miss (red). Too many red bars. "Your 95% CI only covers the truth 82% of the time."

Destroy: Slider controls heteroscedasticity strength. Coverage counter updates live.

Repair: Toggle: classical SEs vs. robust (White/Huber-White) SEs. Coverage snaps to 95% with robust SEs.

Limit: Sample size slider. At n=20, robust SEs have worse coverage than classical (sandwich estimator biased in small samples). Student finds the crossover point.

Disaster sidebar: Google Flu Trends. Autocorrelation inflated confidence. Effective sample size formula: $n(1-\rho)/(1+\rho)$. With $\rho=0.8$ and $n=100$, effective $n\approx11$.

Bridge: "Your standard errors are correct now. But what does 'significant' actually mean?"

8.3 Notebook 3: Why Your Significance Is Wrong

Core concept: Statistical significance vs. effect size and power

Core message: "With enough data, everything is statistically significant. Your shoe size predicts your income. The question is never 'is this significant?' — it's 'is this big enough to matter?'"

Estimated time: 30 minutes

File: 03_significance.ipynb

Trap: 50,000 observations. Everything significant at $p<0.001$. Checkboxes ask: "Which variables have a meaningful effect on revenue?" Student selects based on p-values.

Reveal: Customer satisfaction has $p<0.001$ but coefficient = $0.003 per point. Economically meaningless. Another variable with $p=0.12$ has coefficient = $47.2.

Destroy: Sample size slider. As n increases, p-values for trivial effects cross 0.05. Coefficient stays flat while p-value plummets. Three-panel display: coefficient, SE, t-stat vs. n.

Repair: Introduce effect size measures and confidence intervals. Live power analysis: slider controls true effect size, shows sample size needed to detect it.

Limit: With small n, genuinely large effects can be insignificant. Second slider shows the Type II error rate.

Disaster sidebar: Genome-wide association studies. 1M tests at $\alpha=0.05 \rightarrow$ 50,000 false positives. Bonferroni correction. Mini-sim: 1M t-tests on pure noise.

Bridge: "You know what's significant and what matters. But is your model even the right shape?"

8.4 Notebook 4: Why Your Model Is Wrong

Core concept: Specification error and extrapolation

Core message: "Your model is an approximation. Inside the range of your data, it might be fine. Outside that range, it's fiction with a confidence interval."

Estimated time: 30 minutes

File: 04_specification.ipynb

Trap: Linear model of revenue on ad_spend. $R^2=0.71$. Ask: "How much revenue would a $2M campaign generate?" Student enters prediction.

Reveal: True relationship is logarithmic. Linear extrapolation overshoots by 3x at $2M. Plot shows divergence beyond data range.

Destroy: Draggable vertical bar representing prediction point. Inside data range: overlap. Outside: wild divergence. Confidence band widens but doesn't contain truth.

Repair: Fit log-transformed model. Extrapolation improves dramatically.

Limit: Even log model fails at extreme extrapolation. Every functional form is a local approximation.

Disaster sidebar: Challenger O-ring disaster. Selection on Y (only analyzed failure flights). Include all flights and the temperature trend is obvious.

Bridge: "Your model is now correctly specified. But how do you know it's not just memorizing the data?"

8.5 Notebook 5: Why Your R^2 Is Lying

Core concept: Overfitting and the bias-variance tradeoff

Core message: "A model that fits your existing data perfectly will predict new data terribly. This is not a paradox — it's memorization vs. learning."

Estimated time: 30 minutes

File: 05_overfitting.ipynb

Trap: Add polynomial terms. R^2 climbs: 0.71 → 0.82 → 0.91 → 0.97. Leaderboard format asks: "Which model wins?" Student picks highest R^2.

Reveal: Train/test split. R^2=0.97 model has negative R^2 on test set. Predicts worse than the mean.

Destroy: Slider: polynomial degree 1–15. Two lines: training R^2 (always rising) and test R^2 (rises then crashes). Crossover visible.

Repair: Cross-validation and adjusted R^2. Toggle: switch from polynomial family to log-linear family. Test R^2 jumps above any polynomial. Right model family > right complexity.

Limit: Slider controls number of model specifications searched. Best CV R^2 inflates with search breadth. Even CV can overfit.

Disaster sidebar: Long-Term Capital Management. $4.6B loss. Overfit to historical market patterns. Mini-sim: degree 1–45 polynomial, training MSE →0, test MSE explodes.

Bridge: "Your model fits well and predicts well. But does increasing X cause Y to increase?"

8.6 Notebook 6: Why Your Causal Claim Is Wrong

Core concept: Endogeneity, reverse causality, and instrumental variables

Core message: "If your input and your output cause each other, regression can't tell you which direction the effect runs."

Estimated time: 30 minutes

File: 06_causation.ipynb

Trap: Student has fixed coefficient, SEs, significance, functional form, model selection. Everything looks solid. Ask: "Does increasing ad spend by $100K cause revenue to increase by [coefficient × 100K]?"

Reveal: Time series animation: revenue and ad spend moving together, but revenue leading. Feedback loop exposed. OLS captures mutual dependence, not causal effect.

Destroy: Simulate A/B test (random assignment). Experimental estimate is smaller than observational. Slider controls endogeneity strength. Gap widens.

Repair: Introduce instrumental variables. Natural experiment: ad prices varied randomly across regions. IV estimate recovers truth. Brief primer: relevance, exclusion, Wald estimator (3 cells max). Foreshadowed in NB1 footnote.

Limit: Slider controls instrument strength. Weak instrument: IV estimate oscillates wildly with enormous CIs. Student finds stability threshold.

Disaster sidebar: Police and crime. Positive OLS correlation. Resolved by electoral cycle instrument. Mini-sim: feedback loop DGP, OLS vs. IV estimates.

Bridge: "You've learned six ways your regression can fail. Now let's see what happens when they all attack at once."

8.7 Notebook 7: The Real World

Core concept: Confronting the limits of regression with real data

Core message: "You can do everything right and still get the wrong answer. Because the problem isn't in your model. It's in your data."

Estimated time: 60 minutes

File: 07_real_world.ipynb

Three-Act Structure

Act 1 — False Confidence: Load the Lalonde observational data (base64-encoded in setup cell). Run OLS of earnings on treatment + controls. Coefficient is negative — training appears to reduce earnings. Student runs full diagnostic suite. Everything looks acceptable. Robust SEs barely change things. VIFs under 3. The model passes every test. Trap: "Does the job training program work? Yes / No / Can't tell."

Act 2 — The Confrontation: Reveal the experimental estimate. Positive and large (~$1,800). Side by side with the negative observational estimate. They don't even have the same sign. Visualization: density plots of pre-treatment characteristics for experimental vs. observational samples. Barely overlap. Slider: reweight using subgroup analysis within bins of prior earnings. Estimate crawls toward experimental benchmark but never reaches it. Readout: "Gap remaining: $___"

Act 3 — Wisdom: Sensitivity analysis contour plot (Cinelli-Hazlett). X-axis: partial R^2 of confounder with treatment. Y-axis: partial R^2 with outcome. Shaded region: combinations that explain away the result. Observed covariates plotted as labeled dots. Interactive: student drags a hypothetical confounder point, bias-adjusted estimate updates in real time. Full autopsy report with four fields. Final cell: side-by-side NB1 and NB7 autopsy reports.

Disaster sidebar: Hormone replacement therapy (expanded from NB1 sidebar). Observational studies vs. Women's Health Initiative RCT.

Bridge: "Is regression hopeless? No. You just need the right design."

8.8 Notebook 8: The Redemption

Core concept: Regression discontinuity design — regression works when the design earns trust

Core message: "Regression works brilliantly when the world gives you a natural experiment. Your job is to find those moments — or create them."

Estimated time: 45 minutes

File: 08_redemption.ipynb

Setup: Scholarship cutoff example. Test scores just above and below a threshold. Students barely above got scholarships; those barely below didn't. Near the cutoff, the two groups are nearly identical in every way.

Visualization: Scatterplot with vertical line at cutoff. Regression lines on each side. Gap at boundary = causal effect. Visible. Credible.

Destroy: Bandwidth slider. Narrow: noisy but unbiased. Wide: smooth but biased. Student watches lines wobble (too narrow) or curve toward each other (too wide). Sweet spot visible.

Repair: The RDD itself is the repair. The regression is trustworthy because the design is trustworthy.

Final autopsy report: Everything is green. Threats minimal. Robustness value high. Side-by-side comparison: NB1 report (red, fragile) vs. NB8 report (green, credible). Same tool. Same estimator. Same code. Different credibility.

Disaster sidebar: Maimonides' Rule study (Angrist & Lavy). Israeli class size discontinuity at 40 students.

> *"You've now seen every way a regression can fail. Most practitioners never learn this. They run regressions that look convincing and draw conclusions that feel solid, without knowing what could go wrong. You're different now. You know the failure modes. You know the diagnostics. You know the repairs and their limits. You know how to write an honest autopsy report. This doesn't mean you should stop using regression. It means you're finally ready to start."*

8.9 Notebook 1.5 (Optional): What "Controlling For" Actually Means

Core concept: The Frisch-Waugh-Lovell theorem

Core message: "Multiple regression doesn't 'hold variables constant.' It strips out what the controls explain about both sides, then measures what's left."

Estimated time: 20 minutes

File: 01b_controlling_for.ipynb

Trap: Show multiple regression output. Ask which interpretation is correct: A) holding staff constant, B) after residualizing both sides on staff, C) both mean the same thing, D) neither. Most pick A or C. Correct: B.

Reveal: FWL procedure manually: residualize Y on X_2, residualize X_1 on X_2, regress residuals. Coefficient matches to 14 decimal places. Side-by-side: original scatterplot vs. partial regression plot.

Interactive: Checkbox list of control variables. For each combination, full regression and FWL shown side by side. Match always holds. Partial regression plot updates dynamically. Bad control (X_3) distorts the residual cloud visibly.

Diagnostic tool: Partial regression plot grid for each predictor. Click on any point to highlight it across all plots. Leverage is coefficient-specific, not model-wide.

9. Real-World Disaster Sidebars

Each sidebar has three elements: narrative (3 paragraphs, no jargon), formal diagnosis (OVB formula or equivalent applied to the specific case, 4–6 lines of LaTeX), and mini-simulation (5–10 lines of code, one plot, one button, runs in <2 seconds).

NB	Sidebar Title	Core Error	Formal Diagnosis
1	The hormone therapy that killed	Omitted variable (healthy-user bias)	$\beta_HRT=+0.5$, $\beta_U=-3.0$, $\rho=0.6 \rightarrow E[\hat{\beta}]=-1.3$ (sign flip)
2	Google Flu Trends	Autocorrelation → wrong SEs	Effective $n = n(1-\rho)/(1+\rho)$. $\rho=0.8$, $n=100 \rightarrow$ eff. $n\approx11$
3	Genome-wide association studies	Multiple testing	1M tests, $\alpha=0.05 \rightarrow$ 50K false positives. Bonferroni: $\alpha/m=5\times10^{-8}$
4	Challenger disaster	Selection on Y / specification error	Regress on failure flights only: no trend. All flights: clear trend.
5	Long-Term Capital Management	Overfitting to historical patterns	Degree 45 polynomial: training MSE≈0, test MSE explodes
6	Police and crime	Reverse causality / endogeneity	OLS: positive. IV (electoral cycle): negative. Different signs.
7	Hormone replacement therapy (expanded)	Selection bias + OVB	Full sensitivity analysis. RV shows fragility.
8	Maimonides' Rule	RDD success story	Class split at 40 students. Effect visible at discontinuity.

Sidebar Layout

Each sidebar is a single cell containing an ipywidgets.Tab with three tabs: "The Story", "The Math", "The Simulation". Collapsed by default using #@title with a provocative label. Sidebar plots use 60% saturation of the main color palette. Mini-simulation runs via a single "Run Simulation" button — no sliders, no interactivity. The sidebar is a closed argument: story says X, math says X, simulation confirms X.

10. Cross-Notebook Systems

10.1 Trap Response Storage

Each notebook stores trap responses to /content/trap_log.json. Key format: "notebook_N_qM". Notebook 8's final cell reads this file and compiles a misconception dashboard showing every wrong answer alongside the correction and the theorem that predicted the error.

10.2 Progressive Autopsy Reports

Notebooks 3–6: two-question lightweight version. Notebooks 7–8: full four-question version with sensitivity analysis. Starting from Notebook 3, a dropdown field asks: "Which real-world disaster is most analogous to your situation?" populated with all sidebars encountered so far. This builds the pattern-matching instinct of experienced practitioners.

10.3 Transition Cards

Each notebook ends with a styled HTML card containing: the lingering question, the next notebook's title, and a one-click "Open in Colab" link. Delivered at the moment of maximum engagement.

10.4 Final Dashboard (Notebook 8, last cell)

Side-by-side comparison of the NB1 autopsy report (red, fragile, uncertain) and the NB8 autopsy report (green, credible, trustworthy). Same tool. Same estimator. Same code. Different credibility. Below: table of all trap responses across all notebooks with corrections. Below that: the closing paragraph. "This doesn't mean you should stop using regression. It means you're finally ready to start."

11. Repository Structure & Distribution

11.1 File Structure

```
regression-autopsy/
├── index.html                  (landing page, GitHub Pages)
├── company_sim.py              (shared module, also embedded in each NB)
├── data/
│   └── lalonde.csv             (also base64 in NB7)
├── notebooks/
│   ├── 01_coefficient.ipynb
│   ├── 01b_controlling_for.ipynb  (optional)
│   ├── 02_uncertainty.ipynb
│   ├── 03_significance.ipynb
│   ├── 04_specification.ipynb
│   ├── 05_overfitting.ipynb
│   ├── 06_causation.ipynb
│   ├── 07_real_world.ipynb
│   └── 08_redemption.ipynb
├── images/
│   └── og_card.png             (social media preview)
└── README.md
```

11.2 Landing Page

Static HTML on GitHub Pages. Dark background, white text, course title in large sans-serif, subtitle hook line. Eight full-width cards for core notebooks with titles and one-sentence descriptions. Optional notebooks as indented half-width cards with a subtle border and "Optional Deep Dive" badge. Each card has an "Open in Colab" badge that links directly to the notebook.

11.3 Social Media Card

og_card.png: dark background, white text, title "Regression Autopsy", hook line "Eight Ways Your Model Is Lying to You", subtle scatterplot with regression line in background. Used for Twitter/LinkedIn/Slack previews.

11.4 Self-Containment Principle

Every notebook is fully self-contained. The CompanySimulator and all utilities are embedded in a collapsed setup cell. No cross-notebook imports. No network dependencies for core functionality. The Lalonde dataset is base64-encoded in NB7. A student on a plane with no internet can run any single notebook.

12. Build Order & Phasing

Phase	Deliverable	Purpose	Validates
1	Notebook 1 (complete)	Minimum viable course	Six-beat structure, trap system, Monte Carlo engine, sliders, sidebar, autopsy
2	Notebooks 2 and 7	Core trio	Modularity (NB2) and capstone with real data (NB7). NB7 tests standalone virality.
3	Notebooks 3, 4, 5, 6	Fill the spine	Faster to build — infrastructure exists from Phase 1
4	NB8, NB1.5, instructor's guide, landing page	Finale and polish	Cross-NB dashboard, accessibility, distribution
5	Open source + blog post	Community	Share NB7 on social media (most self-contained/provocative)

Phase 1 produces a complete, shareable artifact. Not a skeleton. A finished notebook a student can open and learn from. That early win sustains momentum through longer phases.

13. Testing & Quality Criteria

13.1 Per-Notebook Testing Checklist

1. Open in fresh Colab runtime with no prior state
2. Setup cell completes in under 30 seconds with visible progress bar
3. Trap widget renders; all options clickable; submit works
4. Attempting to run reveal cell without submitting triggers gate warning
5. Reveal renders with correct tailored response based on trap answer
6. Monte Carlo animation plays smoothly with no frame drops
7. Interactive sliders respond in under 100ms at all positions
8. Theoretical predictions match simulated results to two decimal places at every slider position
9. Repair toggle produces visible, immediate change in visualization
10. Limit slider finds the boundary where repair fails
11. Sidebar tabs render; simulation button produces correct output
12. Autopsy widget accepts input and stores to JSON
13. Transition card links to next notebook correctly
14. Total runtime first-to-last cell: under 3 minutes including precomputation
15. Total memory usage: under 500MB on free Colab tier
16. All plots readable when converted to grayscale
17. All interactive elements have text alternatives (dynamic titles)

13.2 Cross-Notebook Testing

18. Trap log JSON accumulates correctly across notebooks
19. NB8 final dashboard correctly displays all previous trap responses
20. Autopsy report dropdown populates with sidebars from all previously completed notebooks
21. Each notebook runs independently without any other notebook having been opened
22. Transition card Colab links open correctly from within Colab

14. Instructor's Guide Outline

14.1 Timing

Notebook	Self-Paced	Live Classroom	Notes
1	45 min	45 min	First notebook; structure is new to students
2–6	20–30 min each	30 min each	Structure familiar by NB3; faster pacing
7	60 min	60 min	Real data requires more discussion
8	30 min	45 min	Including final reflection and dashboard review
Total	~5 hours	~6 hours	One-day workshop or two-week module

14.2 Suggested Syllabi

Full sequence (8 notebooks): Notebooks 1 → 2 → 3 → 4 → 5 → 6 → 7 → 8. Optional 1.5 between 1 and 2.

Essentials track (4 notebooks): 1 → 2 → 5 → 7. Covers coefficient bias, uncertainty, overfitting, real-world confrontation.

Executive briefing (2 notebooks): 1 → 7. "Your coefficient is wrong" and "you can't always fix it."

14.3 Discussion Facilitation

In a classroom, the instructor should pause at each trap and facilitate public commitment before the reveal. Suggested prompts provided for each notebook. The public commitment makes the correction more powerful in group settings.

15. Accessibility Requirements

- Every color choice paired with secondary visual channel (line style, marker shape). See Visual Design System §4.2.
- All plots must be readable when converted to grayscale. Test every visualization in grayscale before release.
- Dynamic titles carry all quantitative information textually, so screen readers can parse the key lesson.
- Slider labels communicate meaning ("Strong negative confounding" / "Strong positive confounding"), not just numbers. Numbers appear in the summary panel.
- Sidebar tab structure navigable by keyboard.
- All images have alt text in markdown cells.
- Font sizes minimum 10pt in all plots (12pt preferred).
- Notebook runs on free Colab tier with no paid runtime or GPU required.

END OF SPECIFICATION

This document contains everything needed to build the complete Regression Autopsy course.

Build Notebook 1 first. Validate the design. Then iterate.